The Quantum Consensus Principle

A Thermodynamic Information Principle for Quantum Measurement

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Emergent Collapse from Large-Deviation Consensus Dynamics in Open Quantum Systems

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Abstract

Standard formulations of quantum measurement do not by themselves provide a universally accepted dynamical account of single-outcome selection. Within standard open-system quantum mechanics, the Quantum Consensus Principle (QCP) proposes a unified framework connecting microscopic detector dynamics, operational outcome statistics, and apparatus dependence.

We present the Quantum Consensus Principle (QCP), a first principles theory in which measurement outcomes emerge from information-theoretic large-deviation selection inside a macroscopic apparatus. Starting from a microscopic Hamiltonian description of system–apparatus–environment dynamics, we derive a deformed POVM

$$E_i = \sum_{j=1}^{d} S_{ij}\,\Pi_j, \qquad S_{ij} \geq 0, \qquad \sum_{i=1}^{d} S_{ij} = 1 \quad \forall j\,.$$

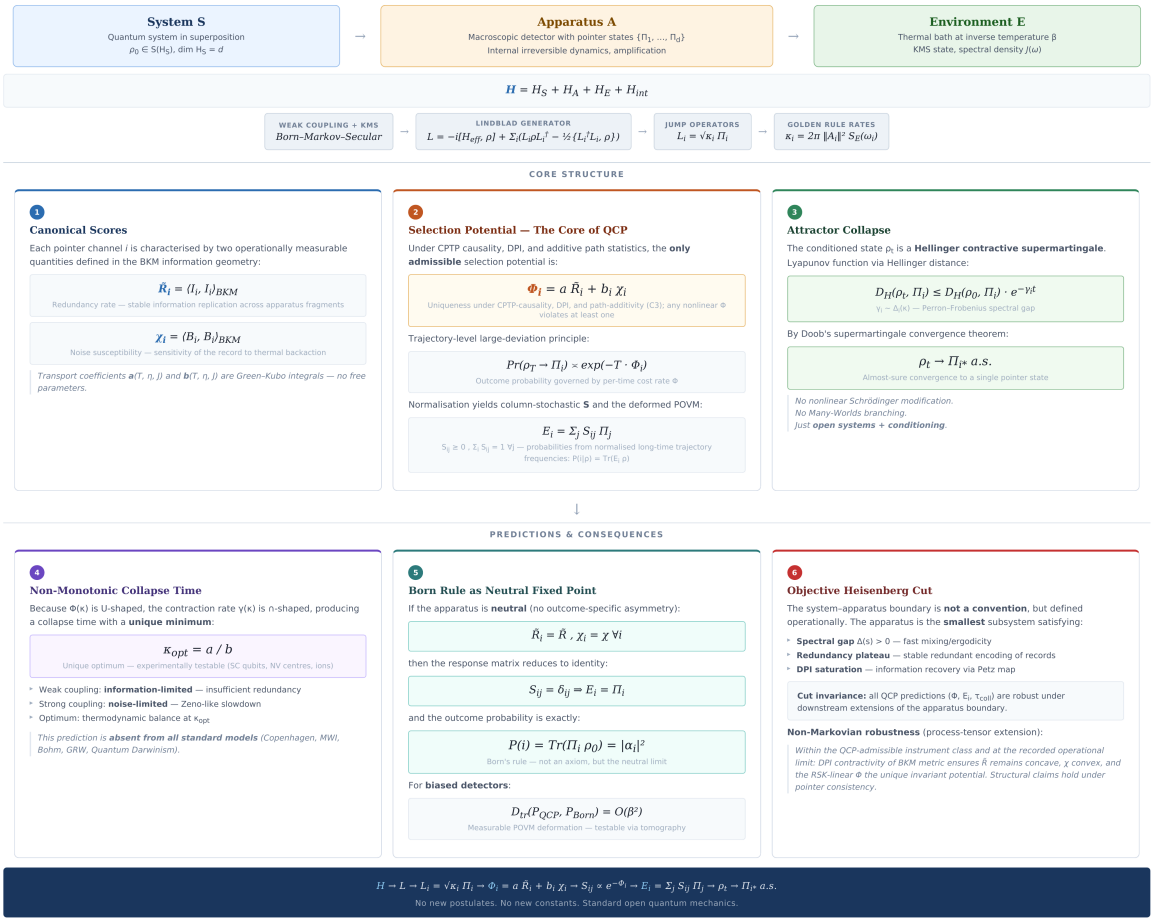
whose weights depend on two canonical functionals of the apparatus response: the redundancy rate $\tilde{R}_i$ and the noise susceptibility χ_i. These quantities, defined in the BKM information geometry of the apparatus, the canonical metric for thermal states, are tied to microscopic Hamiltonian parameters via Green-Kubo transport coefficients $a(T,\eta,J)$ and $b(T,\eta,J)$.

Within the admissible class specified in the companion Supplement causal CPTP dynamics on process-tensor path space together with DPI monotonicity, exact concatenation additivity, convexity/lower semicontinuity, and the stated KMS/regularity assumptions the effective selection functional is unique up to affine gauge and, after calibration of the canonical sufficient statistics, takes the linear form $\Phi_i = a\tilde{R}_i + b_i\chi_i$. Competing nonlinear selectors are excluded only within that explicit admissible class.

We establish rigorously that the conditioned system state ρ_t forms a Hellinger contractive supermartingale, converging almost surely to a unique pointer state Π_{i^*}. Outcome probabilities follow from a path space large-deviation principle, reproducing Born's rule in the apparatus neutral limit and predicting measurable deviations for biased apparatuses. QCP yields a non-monotonic collapse timescale $\tau_{\text{coll}}(\kappa)$, with a unique optimal measurement strength $\kappa_{\text{opt}} = a/b$. These results identify quantum measurement as a thermodynamic consensus phenomenon, deriving collapse, outcome statistics and apparatus dependence directly from Hamiltonian physics without modifying quantum mechanics.

1 | Introduction

Quantum measurement presents an unresolved structural gap: the unitary Liouvillian dynamics of quantum systems does not produce single outcomes, yet experiments do. Projection postulates1–3 formalize collapse without mechanism; Everettian branches[4] preserve unitarity but at the cost of ontic multiplicity. Objective collapse theories5–7 modify the Schrödinger equation, and Bohmian mechanics8 introduces hidden configuration space trajectories. None explain how real detectors amplify microscopic information into macroscopic consensus. Open quantum systems theory9–12 provides precise dynamical tools, but no existing framework derives the emergence of a unique pointer state, nor the Born rule, nor the apparatus dependence of measurement statistics. Here we introduce the Quantum Consensus Principle (QCP): a universal information-theoretic mechanism by which measurement outcomes arise from large-deviation consensus selection in macroscopic detectors. QCP produces a mathematically rigorous collapse, quantitative timescales, auditability from microscopic Hamiltonians, and experimentally accessible deviations from Born statistics. Detailed derivations and mathematical proofs are provided in the Supplementary. A separate companion note provides the complementary necessity statement that, under explicit bare recorded unitary hypotheses, deterministic state-only one-world outcome selection is excluded on the reachable recorded state space.



Comparison of QCP with Prominent Approaches to the Quantum Measurement Problem

The following table summarizes the conceptual and operational differences between the Quantum Consensus Principle (QCP) and major established frameworks addressing the quantum measurement problem. The comparison emphasizes

I. the mechanism that selects a measurement outcome,

II. how (or whether) the Born rule arises,

III. the existence of experimentally distinguishable predictions, and

IV. the theoretical assumptions required by each framework.

Table I | Side-by-side conceptual comparison of quantum measurement frameworks

Framework	Outcome selection	Born rule	Distinctive predictions	Assumptions
Copenhagen	Postulated collapse	Axiom	None	Classical observer cut
Many-Worlds	None (all branches persist)	Derived (symmetry / decision-theoretic)	None beyond standard QM	Ontology of branches
Deutsch-Wallace	Rational choice on Everett branches	Derived (decision-theoretic)	None beyond standard QM	Everett ontology + rationality axioms
Bohmian mechanics	Deterministic pilot-wave dynamics	Reproduced	Nonlocal hidden-variable trajectories	Additional particle ontology
GRW / objective collapse	Spontaneous localization	Modified (non-Born tails)	Heating, localization events	New fundamental constants
Quantum Darwinism	Redundant environment encoding	Assumed	Spectrum-broadcast structure	Standard decoherence
QCP	Thermodynamic attractor selection	Derived (martingale collapse)	Apparatus-dependent POVM deformation	Standard open quantum dynamics

QCP vs. Quantum Darwinism

Quantum Darwinism explains the emergence of redundancy and objective records, but the Born weights are assumed as part of the decoherence initialization. QCP differs in two essential respects:

1. Selection mechanism:
   QCP provides a concrete dynamical selection rule attractor competition governed by the information-thermodynamic rate functional Φ_i^*. Only the thermodynamically stable outcome survives in the long time limit.
2. Probability assignment:
   QCP derives the operational POVM weights directly from its stochastic dynamics. In the neutral device limit, the martingale theorem recovers the standard Born rule; for general apparatuses, device parameters $(\kappa_i, \chi_i, \tilde{R}_i)$ induce controlled POVM deformations.

QCP vs. Many-Worlds

Many-Worlds avoids collapse by asserting that all branches are equally real. QCP instead provides single-outcome selection without invoking additional ontology:

- Losing branches decay as $\exp(-\Phi_i^*)$ under the large-deviation dynamics.
- There is no multiplicity of coexisting macroscopic worlds.

QCP vs. GRW

GRW modifies the Schrödinger equation by introducing new constants and spontaneous localization events. QCP remains entirely within standard open-system quantum mechanics:

- No new stochastic constants or universal length/time scales.
- Collapse emerges from the interplay of Lindblad dynamics, information rates, and Doob-transformed attractors.

QCP vs. Bohmian Mechanics

Bohmian mechanics introduces additional particle ontology and relies on initial $|\psi|^2$-equilibrium to reproduce Born weights. QCP preserves the Hilbert-space ontology and obtains measurement probabilities from:

- stochastic trajectories,
- martingale collapse,
- and the device specific large-deviation structure Φ_i^*.

Summary

This comparison highlights that the Quantum Consensus Principle (QCP) offers a genuine dynamical mechanism for outcome selection and a first-principles derivation of measurement probabilities. It predicts apparatus-dependent and experimentally accessible deviations while remaining entirely within the framework of standard open-system quantum mechanics. Importantly, QCP achieves this without introducing any new constants, stochastic parameters, or additional ontological entities beyond those already present in quantum theory.

Scope and logical status.

The present main paper states the physical architecture of QCP and its front-end consequences in the Markovian pointer-preserving regime. The full admissibility, uniqueness, calibration, process-tensor extension, and counterexample-exclusion proofs are carried by the companion Supplement and notes. Accordingly, the strongest claims made here are to be read conditional on those standing assumptions and calibration conditions, rather than as unrestricted statements about arbitrary detector models.

2 | Microscopic Model and Conditional Dynamics

Consider system S, apparatus A, and environment E governed by $H = H_S + H_A + H_E + H_{\text{int}}$.

Under standard weak coupling, rotating wave, and coarse graining approximations, the conditional dynamics of the system is governed by a stochastic master equation

$$\mathrm{d}\rho_t = \mathscr{L}[\rho_t]\,\mathrm{d}t + \sum_i \left[\mathscr{J}_i(\rho_t) - \rho_t\,\mathrm{Tr}(\mathscr{J}_i(\rho_t))\right]\mathrm{d}N_i(t),$$

with Lindblad generator

$$\mathscr{L}[\rho] = -\,i[H_{\text{eff}}, \rho] + \sum_i \left(L_i \rho L_i^\dagger - \tfrac{1}{2}\{L_i^\dagger L_i, \rho\}\right),$$

and jump operators $L_i = \sqrt{\kappa_i}\,\Pi_i$. The microscopic rates $\kappa_i = 2\pi\,\|A_i\|^2\,S_E(\omega_i)$ follow directly from the Golden Rule and the bath spectral density $S_E(\omega)$; upward/downward rates for non-QND couplings follow KMS relations. QCP uses no phenomenological parameters: all apparatus asymmetries enter through κ_i and through correlation functions defining the canonical scores below.

3 | Canonical Scores: Redundancy and Susceptibility

Macroscopic detectors amplify microscopic alternatives via large networks of degrees of freedom. For each output channel i, QCP defines two canonical scores:

Redundancy rate $\tilde{R}_i$

The asymptotic rate at which stable, distinguishable information about outcome i is produced in the apparatus. In the formalism, $\tilde{R}_i = \langle I_i, I_i\rangle_{\text{BKM}}$, a BKM-metric quadratic form of the apparatus information current operator I_i.

Noise susceptibility χ_i

The sensitivity of the apparatus's record to thermal and entanglement induced noise. $\chi_i = \langle B_i, B_i\rangle_{\text{BKM}}$, with B_i the corresponding noise current.

Green-Kubo transport coefficients

The coefficients $a(T, \eta, J)$ and $b(T, \eta, J)$ entering the QCP potential are given by linear response integrals:

$$a = \int_0^\infty \langle I_i(t) I_i(0)\rangle\,\mathrm{d}t, \qquad b = \int_0^\infty \langle B_i(t) B_i(0)\rangle\,\mathrm{d}t.$$

These link Φ directly to the microscopic Hamiltonian. Crucially, the scores $(\tilde{R}_i, \chi_i)$ themselves are static information geometric quantities, while a,b quantify dynamical transport.

Notation convention.

We use the standard Lindblad normalization $L_i = \sqrt{\kappa_i}\Pi_i$, where \kappa_i are Golden-Rule jump rates determined by microscopic couplings and bath spectra. For channel-resolved outcome competition we write $\Phi_i = a\tilde{R}_i + b_i\chi_i$. In symmetric or one-parameter calibration scans we use the reduced notation $b_i \equiv b$ and define the scalar scan functional $\Phi_{\text{scan}}(\kappa) := b\,\chi(\kappa) - a\,\tilde{R}(\kappa)$, which is a coupling-tuning functional and not a redefinition of the channel selector Φ_i.

4 | The Quantum Consensus Potential

A central theorem in the companion Supplement establishes the following main-paper form of the selector result.

Theorem (Admissible linear selector, main-paper form). Within the explicit admissible class defined in the companion Supplement causal CPTP/process-tensor dynamics, DPI monotonicity, exact concatenation additivity, convexity/lower semicontinuity, and the stated support/KMS regularity assumptions the QCP selection functional is unique up to affine gauge. After identification of the two canonical sufficient statistics $\tilde{R}_i$ and χ_i, this yields the calibrated linear form $\Phi_i = a\,\tilde{R}_i + b_i\chi_i$.

The stronger path-space uniqueness and no-competing-divergence statements are proved in the Supplement. Competing nonlinear selectors are excluded within the above admissible class; outside that class, no such claim is made in the present paper.When scanning over measurement strength κ, the effective potential becomes

$$\Phi(\kappa) = b\,\chi(\kappa) - a\,\tilde{R}(\kappa),$$

whose geometric constraints enforce a U-shaped profile with a unique minimum $\kappa_{\text{opt}} = \dfrac{a}{b}$. Within the calibrated scan family considered here, this structure underlies the non-monotonic collapse time.

The primary QCP selector is Φ_i. The channel-wise entropy-production rate $\sigma^{(i)}$ serves as a complementary thermodynamic diagnostic. In general, the minimizers of $\sigma^{(i)}$ and Φ_i need not coincide; equality is asserted only on calibrated apparatus classes for which the pairwise order condition (*) of the companion Supplement is verified.

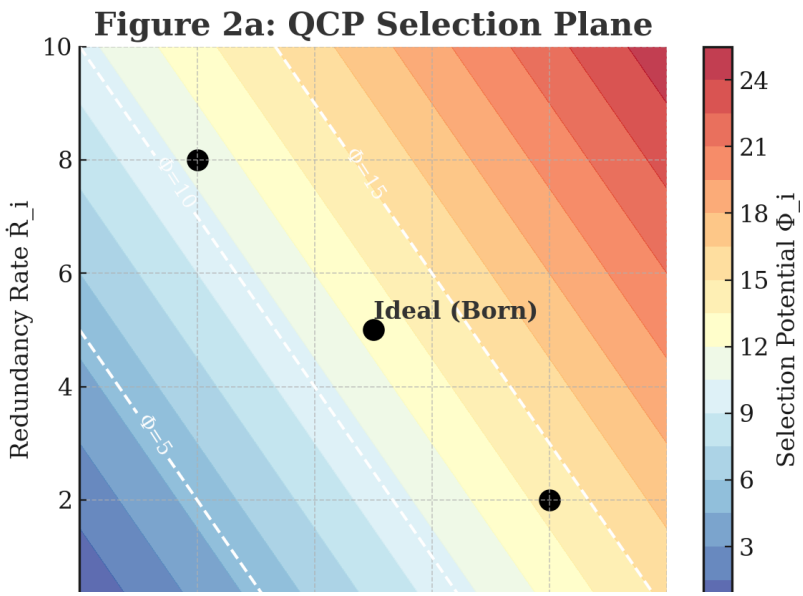


FIG.2a Conceptual selection landscape of the Quantum Consensus Principle (QCP). The selection potential $\Phi_i = a\,\tilde{R}_i + b\,\chi_i$ is plotted over the redundancy susceptibility plane. Color contours indicate constant Φ_i values (iso-potential lines). The central "Ideal (Born)" point corresponds to the thermodynamically neutral apparatus limit $\tilde{R}_i = R, \chi_i = \chi$, where Born's rule is recovered. Deviations from neutrality produce measurable shifts $\Delta\Phi_i \propto O(\beta^2)$, predicting testable deviations from Born level statistics. This figure is conceptual and illustrates the theoretical geometry of the QCP selection mechanism.

5 | Large-Deviation Derivation of Outcome Probabilities

Let ρ_t denote the conditional state trajectory. Using a Doob h-transform of the path measure, we prove the trajectory level large-deviation principle

$$\Pr(\rho_T \to \Pi_i) \asymp \exp[-T\,\Phi_i].$$

Normalization defines a column-stochastic response matrix S and the QCP-POVM

$$E_i := \sum_{j=1}^{d} S_{ij}\Pi_j, \qquad S_{ij} \geq 0, \qquad \sum_{i=1}^{d} S_{ij} = 1\ \forall j, \qquad P(i\,|\,\rho) = \mathrm{Tr}(E_i\rho)\,.$$

Operational probability law and neutral limit.

In QCP, outcome probabilities are defined operationally as asymptotic frequencies of stochastic trajectories. For a general calibrated instrument, the exact output law is

$$P(I^* = i) = \mathrm{Tr}(E_i\rho_0), \qquad E_i = \sum_j S_{ij}\Pi_j\ ,$$

with S column-stochastic. The neutral apparatus is defined operationally by $S = I$, equivalently $E_i = \Pi_i$. In that neutral case, the posterior process $M_i(t) := \mathrm{Tr}(\Pi_i\rho_t)$ is a bounded martingale, and together with almost-sure collapse to pointer sectors this yields

$$P(I^* = i) = \mathrm{Tr}(\Pi_i\rho_0).$$

Thus the Born law is the exact trajectory-frequency law of the neutral instrument, not an independent probability postulate. Away from neutrality, first-order deviations from Born are stated only for smooth response-matrix families $S(\varepsilon)$.

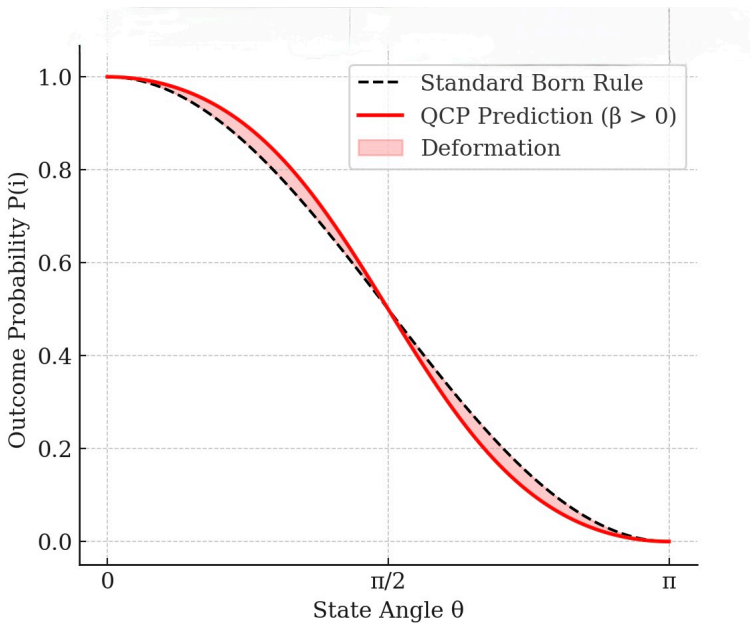


FIG.4a Conceptual prediction of deviations from the Born rule in the Quantum Consensus Principle (QCP). The QCP probability distribution $P_{\mathrm{QCP}}(\theta)$ (red) deviates slightly from the standard Born rule (black dashed) in the presence of non-neutral apparatus conditions $(\beta > 0)$. These deviations reflect thermodynamic asymmetries in redundancy and susceptibility, $\tilde{R}_i \neq R, \chi_i \neq \chi$. The Born rule is recovered as the neutral fixed point $\beta \to 0$. This figure is conceptual and illustrates the measurable deformation predicted by the QCP framework.

6 | Emergent Collapse Dynamics

The stochastic process $\{\rho_t\}$ is a supermartingale under the natural filtration generated by measurement events. We construct a Lyapunov function given by the Hellinger distance

$$H_i(t) = D_H\big(\rho_t, \Pi_i\big),$$

and prove exponential contraction:

$$H_i(t) \le H_i(0)\, e^{-\gamma_i(\kappa)t}, \qquad \gamma_i(\kappa) \sim \Delta_i(\kappa).$$

Here $\Delta_i(\kappa)$ is the Perron–Frobenius spectral gap of the Doob-transformed generator. As shown in the Supplement, the contraction rate is a monotone decreasing function of the potential: smaller $\Phi \to$ larger $\gamma \to$ faster collapse.

Theorem (Consensus collapse).

$\rho_t \to \Pi_{i*}$ almost surely. In the pointer-preserving Markovian regime treated explicitly in the present main text, and under the standing regularity assumptions detailed in the companion Supplement, the conditioned state process admits a Hellinger-type Lyapunov functional and converges almost surely to a pointer projector. The quoted exponential rate bound is proved in the weak-coupling / band-limited Doob-tilted setting used there. More general extensions preserve the qualitative selection architecture, while quantitative contraction-rate statements remain tied to the stated regularity hypotheses..

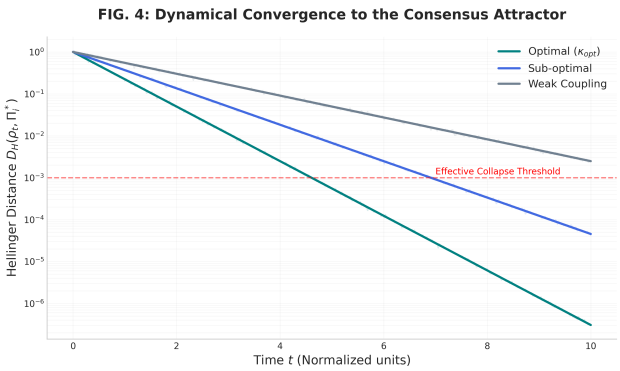


FIG.4 Conceptual dynamical convergence to the consensus attractor in the Quantum Consensus Principle (QCP). The Hellinger distance $D_H(\rho_t, \Pi_i^*)$ decays exponentially, indicating convergence of the conditional quantum state ρ_t toward the consensus attractor Π_i^*. Different curves correspond to weak, sub-optimal, and optimal coupling strengths κ. The red dashed line denotes the effective collapse threshold beyond which the system is considered classically stable. The fastest convergence occurs at the optimal coupling $\kappa_{\text{opt}} = a/b$. This figure is conceptual and illustrates the stochastic relaxation predicted by the QCP framework.

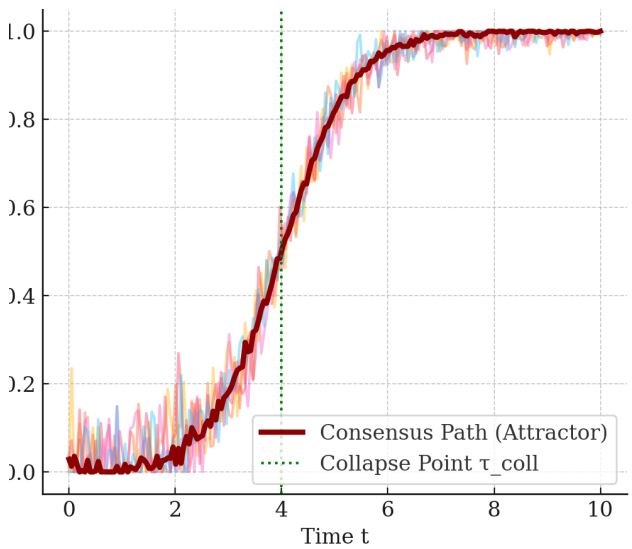


FIG.5 | Conceptual time evolution of consensus formation in the Quantum Consensus Principle (QCP). The consensus weight $w_i(t)$ evolves toward unity as the conditional quantum state ρ_t converges to the attractor Π_i^*. Colored lines represent stochastic trajectories, while the thick red line denotes the ensemble average. The vertical green line marks the collapse point τ_{coll}, corresponding to the completion of consensus formation. This figure is conceptual and illustrates the stochastic path convergence predicted by the QCP framework. We track the pointer probability via the consensus weight $w_i(t) := \text{Tr}[\Pi_i \rho(t)]$ (see Supplement Thm. 5.2 for the posterior martingale $M_i(t)$).

Non-monotonic collapse time

$$\tau_{\text{coll}}(\kappa) \sim \frac{1}{\gamma_{\min}(\kappa)}.$$

Because $\Phi(\varkappa)$ is U-shaped, $\gamma(\varkappa)$ is ∩-shaped, and $\tau_{\text{coll}}(\kappa)$ has a unique minimum at $\kappa_{\text{opt}} = a/b$. This prediction is absent from all standard models.

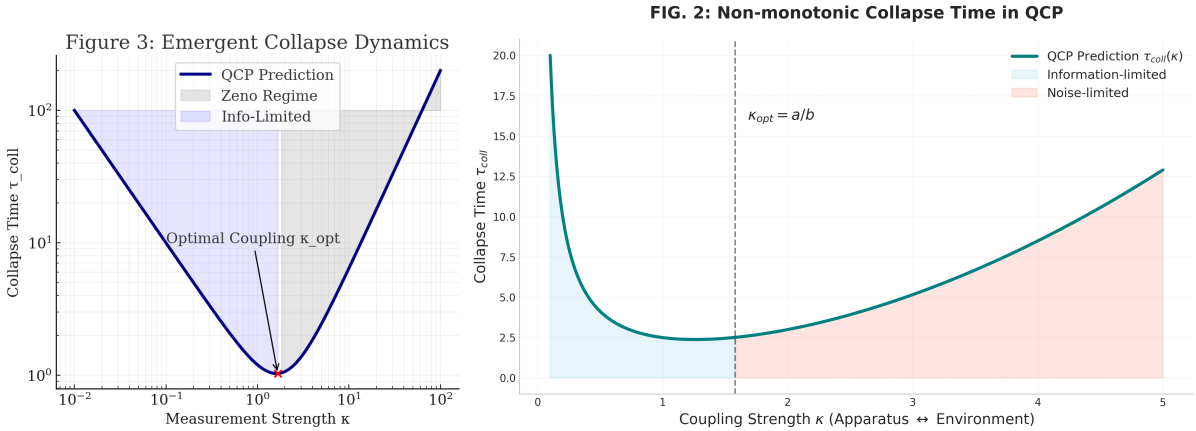


FIG.3 Conceptual prediction and emergent scaling of collapse time dynamics in the Quantum Consensus Principle (QCP). The collapse time τ_{coll} predicted by QCP exhibits a characteristic non-monotonic dependence on the apparatus environment coupling strength κ. For weak coupling (blue region), the process is information limited due to insufficient redundancy amplification, whereas for strong coupling (red region) it becomes noise-limited owing to excessive thermal backaction. The minimum at $\kappa_{\text{opt}} = a/b$ identifies the thermodynamically optimal consensus rate determined by the QCP selection potential $\Phi_i = a\,\tilde{R}_i + b\,\chi_i$.

FIG.2 | presents the thermodynamic representation of this balance, highlighting the interplay between redundancy and susceptibility, while Figure 3 shows the same relationship on a log–log scale, emphasizing the emergent scaling across several orders of magnitude and the transition between the information limited and Zeno regimes. Together, these figures are conceptual and illustrate the universal structure of collapse dynamics and scaling predicted by the QCP framework.

7 | Non-Markovian Robustness

Using the Markovian pointer-preserving regime as the reference front-end model. The companion Supplement formulates the corresponding path-space uniqueness, POVM validity, and no-signalling statements on process-tensor path space under explicit causality, DPI, additivity, support, and KMS/ regularity assumptions. Accordingly, robustness beyond Markov/QND is claimed here only in that conditional sense.The formal derivation of the process-tensor formulation and the proof of invariance of the RSK-linear potential under non-Markovian dynamics are given in Supplementary Section 2.12.

8 | Predictions and Falsifiability

QCP makes several experimentally accessible predictions:

Non-monotonic collapse rate

In continuous-measurement platforms (superconducting qubits, NV centers, trapped ions), varying measurement strength $\varkappa$ must yield $\tau(\kappa)$ U-shaped, with a device specific optimum.

Born deviations for biased apparatus

For asymmetrically structured detectors,

$$D_{\mathrm{tr}}(P_{\mathrm{QCP}}, P_{\mathrm{Born}}) = O(\beta^2),$$

with β an apparatus bias parameter. Modern POVM tomography can resolve this.

Microscopic auditability

All parameters of the Quantum Consensus Principle (QCP) are operationally measurable and thus experimentally verifiable.

The microscopic coupling strengths κ_i can be determined from spectroscopy of the environmental response, while the transport coefficients a and b are obtained from Green–Kubo integrals characterizing energy and information flow.

The redundancy rates $\tilde{R}_i$ and noise susceptibilities χ_i can be extracted through partial monitoring of the environment, providing direct access to the dynamics of information amplification.

Finally, the effective measurement operators E_i are reconstructable via multi-strength quantum tomography, allowing full empirical recovery of the POVM structure predicted by QCP.

Together, these measurable quantities render the Quantum Consensus Principle a fully falsifiable physical theory, in which every theoretical parameter corresponds to an experimentally accessible observable.

9 | Discussion

QCP reconceives measurement as an information driven nonequilibrium phase selection process. Collapse is not an exception inserted into the formalism; it is the inevitable attractor of redundancy amplifying dynamics. The Born rule becomes a statement of apparatus neutrality, not a primitive axiom. The theory offers:

- no new ontology,
- no hidden variables,
- no nonlinear Schrödinger corrections,
- no branching universes.

Instead, outcome selection results from thermodynamic consensus, governed by universal constraints on CPTP maps, DPI, and information geometry.

The formalism unifies quantum measurement with statistical physics: macroscopic measurement devices act as large-deviation engines, computing the most stable macroscopic alternative.

10 | Conclusion

We have presented the Quantum Consensus Principle (QCP), a first-principles framework that derives quantum measurement as an emergent thermodynamic phenomenon rather than a postulated rule.

Starting from microscopic Hamiltonian dynamics and extending through information-geometric and path-space large-deviation formalisms, QCP provides a unified and quantitatively predictive description of measurement. In the Markovian pointer-preserving regime emphasized in the present main text, and under the standing assumptions carried by the companion materials, QCP yields an almost-sure collapse architecture together with an apparatus-dependent POVM description and a non-monotonic calibrated collapse-time prediction. In the neutral instrument limit $S = I$, the Born law is recovered as the exact operational trajectory-frequency law; away from neutrality, calibrated response matrices generate controlled POVM-level deviations that are, in principle, experimentally testable. By grounding outcome selection in the universal thermodynamics of information production, QCP transforms the measurement problem from an interpretational postulate into a derived physical process. Comprehensive mathematical derivations, calibration protocols, and formal proofs supporting these results are provided in the Supplementary Information accompanying this article.

A separate companion note supplies a narrower necessity statement than a universal no-go: under bare recorded unitary dynamics, explicit record hypotheses, support faithfulness, and reachable relabeling covariance, no deterministic state-only single-outcome selector exists on the reachable recorded state space. This closes one direct state-only route within the stated formal setting, while the positive uniqueness of the admissible path-space selector remains the independent content of the companion Supplement.

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